

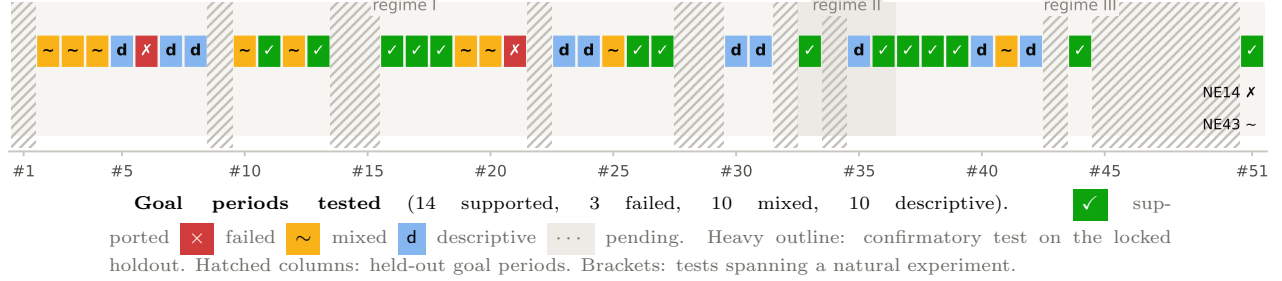
H86: Taylor's law as a shared-field gauge

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 45% Faithfulness 2/5 Usefulness 2.5/5

Bottom line. Taylor's c measures clock regularity, not the field; the pair-covariance $c_{\times}$ measures the field, which is small and mostly the scheduler.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

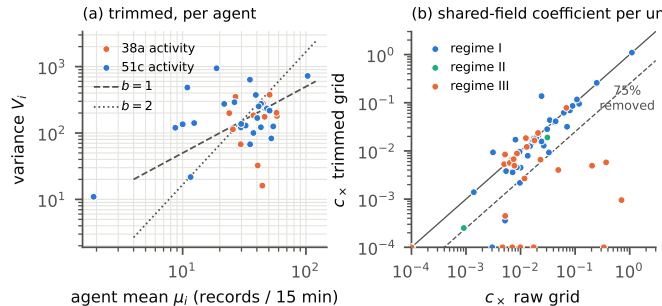
The test asks whether Taylor's law gives a one-number gauge of the shared field. Per period unit, each agent's activity is counted in 15-minute bins, on the raw day window and on DQ8's all-present (trimmed) window. The HH's coefficient c_T (variance = $a\mu + c_T\mu^2$ across agents) fails: Taylor's exponent is $b \approx 0.85$ and $c_T \leq 0$ in most of 70 units. The pair-covariance coefficient $c_{\times}$ works as a gauge. It is small (0.016 raw, 0.008 trimmed), and the shared share of excess variance is 0.08. Trimming removes 77% of it in regime III, matching H38's 70–80%. The gauge is now a per-unit column. The holdout script exists and is not run.

PHYSICAL MODEL

Fluctuation scaling (model 14) with a multiplicative field $h(t)$ (model 02). Agent i 's count in bin t is Poisson with rate $\mu_i \xi_t \zeta_{it}$. ξ_t is a field shared by all present agents (scheduler, kickoff, operator) with variance $c_{\times}$. ζ_{it} is the agent's private burstiness with variance s_i . $c_{\times}$ is read from pair covariances and needs no spread in μ . φ is the shared share of super-Poisson variance.

$$V_i = \mu_i + [c_{\times} + s_i(1 + c_{\times})]\mu_i^2, \quad \text{Cov}(y_i, y_j) = c_{\times} \mu_i \mu_j$$

$$c_{\times} = \frac{\sum_{i < j} \text{Cov}_{ij}}{\sum_{i < j} \mu_i \mu_j}, \quad \varphi = \frac{c_{\times} \sum_i \mu_i^2}{\sum_i (V_i - \mu_i)}, \quad r_{\times} = 1 - \frac{c_{\times}^{\text{trim}}}{c_{\times}^{\text{raw}}}$$



Observables. (a) Variance against mean per agent, trimmed activity, units 38a and 51c. The points do not follow $b = 2$; faster agents are more regular. (b) Shared-field coefficient per unit, raw against trimmed grid. Regime-III units fall toward the 75%-removed line; regime-I units sit on the diagonal.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

Report `taylor_c_shared` on `activity_trim` (and `activity_raw`) from `per_period_estimates` next to any synchrony or influence claim. A trimmed $c_{\times}$ near 0.008 with $\varphi < 0.1$ means the swarm's activity fluctuations are about 90% private. Do not use Taylor's b or c_T as field measures in a clocked swarm: they track clock regularity. A raw $c_{\times}$ far above the trimmed one (e.g. 2.2 on 2026-03-31) flags a village-off gap inside the day window. Talk shares about 2.6 times more of its variance than activity (φ 0.21 vs 0.08).

RESULTS VS PREDICTIONS

FAITHFULNESS SCORECARD

Prediction	Observed	Verdict
P1 raw $b \approx 2$, $c_T > 0$	b 0.85 [0.21, 1.21]; $c_T > 0$ in 31%	fail
P2 trimmed $b \leq 1.3$	0.82 (raw already < 1)	trivial
P3 III: $r_x \in [0.6, 0.9]$; $I < 0.4$	III 0.77 (13 units); I 0.06 (25)	pass (c_x)
P4 kill: III $\varphi_{\text{trim}} \geq 0.5$	φ 0.04; shift null $p < .05$ in 38%	not fired
P5 call clock lowers b	lower in 58% (need $\geq 67\%$)	fail
P6 commits; specificity	b 1.26; ρ 0.14 ($n = 6$)	b pass
P7 r_x vs H38 f_{scaffold}	ρ 0.19 (13 units)	fail
P8 talk $\varphi >$ activity	77% of 64 units; 0.21 vs 0.08	pass
N1 NE14 raw jump	one off-gap day; trimmed +0.001	fail
N2 NE43 operator stops	c_x unchanged; c_T no rise	mixed

A mapping 1 F identifiability 2
 B assumptions 1 G ground truth 1
 C adequacy 1 H comparative 1
 D unfitted predictions 1 I transfer 1
 E interventional 1
 Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writup/paper.tex`).

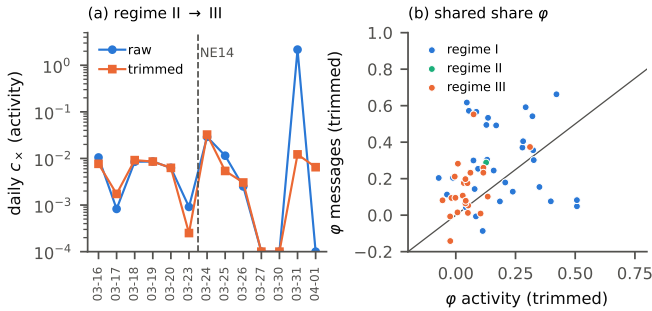
CAVEATS

Across-agent fits have 4–32 points and a narrow mean range in regime III, so unit-level b is unstable; the card uses medians. Day-bootstrap CIs under-cover (0.7–0.8 in synthetic runs). c_x counts equal-time coupling as field. Statistics depend on the 15-minute bin. The commit gauge exists in 13 units, so the kickoff-specificity test has 6 goal periods. The regularity reading of $b < 1$ is post hoc.

ROUND 2

Fit a clock-aware Taylor law (renewal model of each agent's calls) and report the residual shared term. Split c_x into equal-time and one-call-lagged parts on the call clock, which separates field from read-out coupling. Use raw minus trimmed c_x as a daily off-gap flag. Run the frozen holdout test after sign-off.

- **Essence.** one number per unit that says how much of the swarm's fluctuation is shared, after the scheduler is removed.
- **H86-R1** Clock-aware Taylor law.
- **H86-R2** Lagged vs equal-time c_x .
- **H86-R3** Off-gap detector.
- **H86-R4** Units 6a and 21b.



Second figure. (a) Daily c_x across NE14 (regime II to III). Raw and trimmed agree except on 2026-03-31, whose window holds a 513-minute all-silent gap. (b) Shared share φ on trimmed grids, messages against activity, one point per unit: talk shares more.

GOAL PERIODS IN DETAIL

G02 ~ mixed
 G03 ~ mixed
 G04 ~ mixed
 G05 d descriptive
 G06 x failed
 G07 d descriptive
 G08 d descriptive
 G10 ~ mixed
 G11 ✓ supported
 G12 ~ mixed

+27 more periods: see the card.